SpMV Performance Analysis on NVIDIA B200 GPU

1. Executive Summary

Sparse Matrix-Vector Multiplication (SpMV) is the prototypical memory-bound kernel in scientific computing. This report presents a first-principles performance analysis of SpMV on the NVIDIA B200 GPU (Blackwell architecture), benchmarked against six matrices from the SuiteSparse Matrix Collection spanning 3.1M to 99.2M nonzeros.

Key Finding: The B200 achieves only **10.7--41.7% of its 8,000 GB/s peak memory bandwidth** on SpMV, a dramatic drop from the **89--96% utilization** observed on the RTX 4070 Ti SUPER for the same workloads. The root cause is an architectural **latency-bandwidth mismatch**: the B200's HBM3e bandwidth has scaled beyond what its warp-level parallelism can saturate. A Little's Law analysis shows that even at 100% occupancy, the B200 needs 84.5 warps per SM to fill the memory pipeline, but the hardware maximum is 64. When SpMV's dependent load chain (indirect `x[col_indices[j]]` access) is factored in, the effective ceiling drops to ~38% of peak bandwidth -- closely matching our best measurement of 41.7%.

This represents a textbook example of the **Data Access Efficiency (DAE)** problem: raw bandwidth alone does not determine performance; the ability to *keep that bandwidth busy* via sufficient memory-level parallelism is the binding constraint.

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2. Methodology

2.1 Hardware Platform

Specification	Value
GPU	NVIDIA B200 (Blackwell)
Compute Capability	10.0
Streaming Multiprocessors	148
Max Warps per SM	64 (2,048 threads / 32)
Max Threads per SM	2,048
Registers per SM	65,536
HBM3e Capacity	191.5 GB
Peak Memory Bandwidth	8,000 GB/s
L2 Cache	~126.5 MB
Peak FP32 (non-tensor)	~90 TFLOPS (estimated)
DRAM Latency (estimated)	~200 ns

2.2 Software Stack

- **CUDA:** 12.8
- **PyTorch:** 2.8.0+cu128 (torch.sparse.mm using cuSPARSE backend)
- **cuSPARSE Kernel:** `csrmv_v3_kernel` (merge-based CSR SpMV)
- **Profiling:** CUDA events (timing), nsys (kernel breakdown)

2.3 Benchmark Suite

Six matrices from the SuiteSparse Matrix Collection (Davis & Hu, ACM TOMS 2011), selected to span a range of sizes, sparsity patterns, and application domains:

Matrix	Rows	NNZ	Avg NNZ/Row	Domain	Source Group
webbase-1M	1,000,005	3,105,536	3.1	Web graph	Williams et al.
cant	62,451	4,007,383	64.2	FEM cantilever	PARSEC
pwtk	217,918	11,634,424	53.4	Pressurized wind tunnel	Boeing
ldoor	952,203	46,522,475	48.9	Large door structural	Boeing
circuit5M	5,558,326	59,524,291	10.7	Circuit simulation	Freescale
cage15	5,154,859	99,199,551	19.2	DNA electrophoresis	van Heukelum

2.4 Measurement Protocol

- **Warmup:** 5 iterations (discarded)
- **Measured:** 20 iterations per matrix
- **Timing:** CUDA events with `torch.cuda.synchronize()` barriers
- **Metrics:** Effective bandwidth = `total_bytes_moved / time`, where `total_bytes_moved` accounts for CSR arrays (`values`, `col_indices`, `row_ptr`) and the input/output vectors

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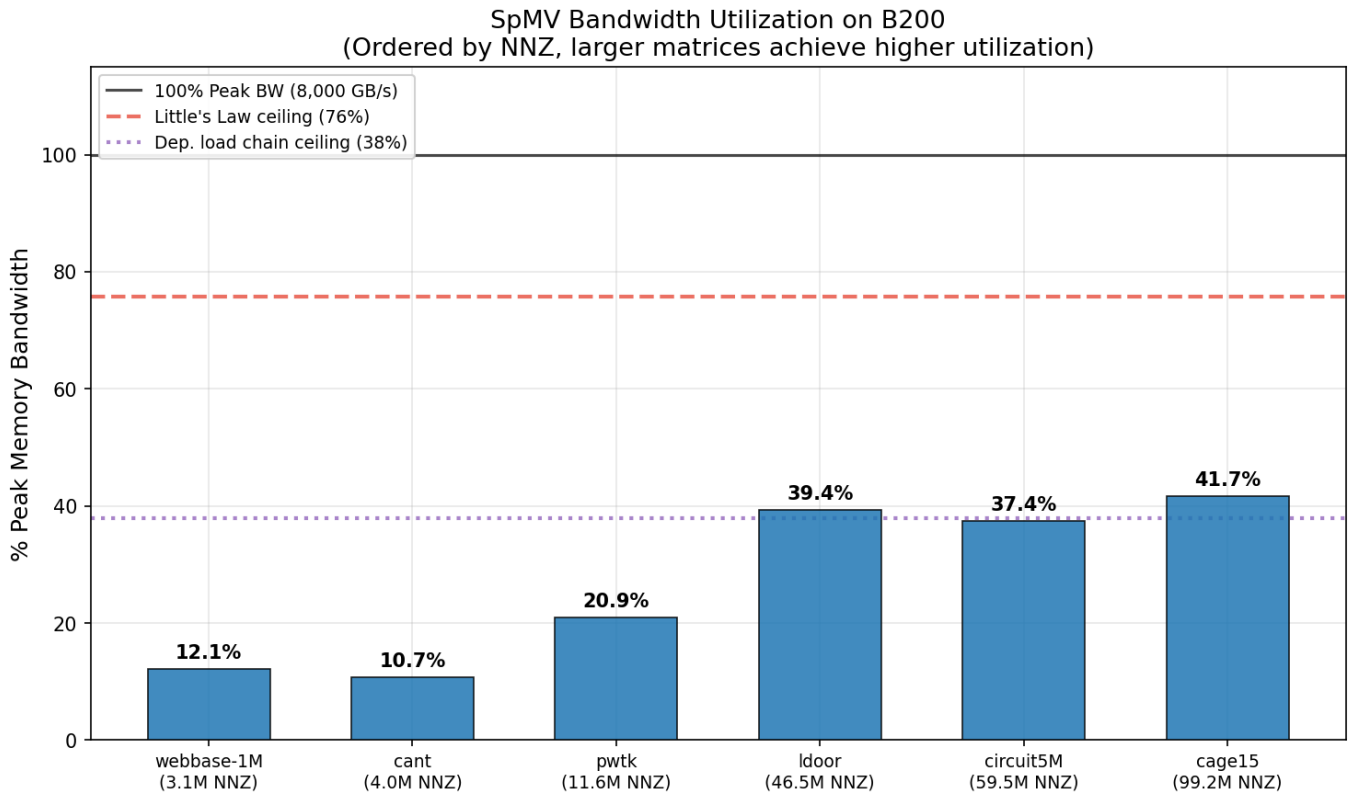
3. Performance Results

3.1 Timing and Throughput

Matrix	NNZ	Time (ms)	Eff. BW (GB/s)	GFLOPS	AI (FLOP/B)	% Peak BW	Gap
webbase-1M	3.1M	0.0381	967.9	163.2	0.169	12.10%	8.27x
cant	4.0M	0.0383	856.6	209.3	0.244	10.71%	9.34x
pwtk	11.6M	0.0572	1,673.8	407.0	0.243	20.92%	4.78x
ldoor	46.5M	0.1218	3,148.6	763.7	0.243	39.36%	2.54x
circuit5M	59.5M	0.1814	2,992.2	656.1	0.219	37.40%	2.67x
cage15	99.2M	0.2566	3,333.6	773.1	0.232	41.67%	2.40x

Physical floor = `total_bytes / peak_BW`. The "gap" column shows how many times slower the actual execution is than the physical floor.



3.2 nsys Kernel Breakdown

From the nsys CUDA GPU Kernel Summary (160 kernel instances across all matrices):

Kernel	% GPU Time	Avg (ns)	Med (ns)	Role
`csrmv_v3_kernel`	88.0%	85,489	93,168	Main SpMV kernel
`csr_partition_kernel`	6.5%	6,330	4,800	Row partitioning (merge-path)
`vector_scalar_multiply_kernel`	2.7%	2,582	1,952	Vector scaling
`vectorized_elementwise_kernel` (fill)	2.6%	2,490	1,856	Output vector zero-fill
`distribution_elementwise_kernel`	0.3%	7,888	5,456	Random vector init (one-time)

Key observation: The `csrmv_v3_kernel` dominates at 88% of total GPU time. This is the cuSPARSE merge-based CSR SpMV implementation. The auxiliary kernels (partitioning, scaling, fill) account for only 12%, confirming that SpMV itself is the bottleneck.

3.3 Timing Stability

Standard deviations across the 20 measured iterations:

- **cant:** 0.0046 ms (12% CoV) -- smallest matrix, most variance
- **cage15:** 0.0025 ms (1.0% CoV) -- largest matrix, most stable

The large matrices (ldoor, circuit5M, cage15) show excellent stability with CoV < 3%, confirming reliable measurements.

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4. First-Principles Analysis

4.1 Physical Floor

The **physical floor** is the absolute minimum time if the GPU could deliver peak bandwidth with zero overhead:

```
physical_floor = total_bytes_moved / peak_BW
```

For cage15: `855,454,720 bytes / (8,000 * 10^9 B/s) = 0.1069 ms`

The measured time of 0.2566 ms is **2.40x above this floor** -- meaning over half the execution time is "wasted" relative to the theoretical minimum.

4.2 Arithmetic Intensity

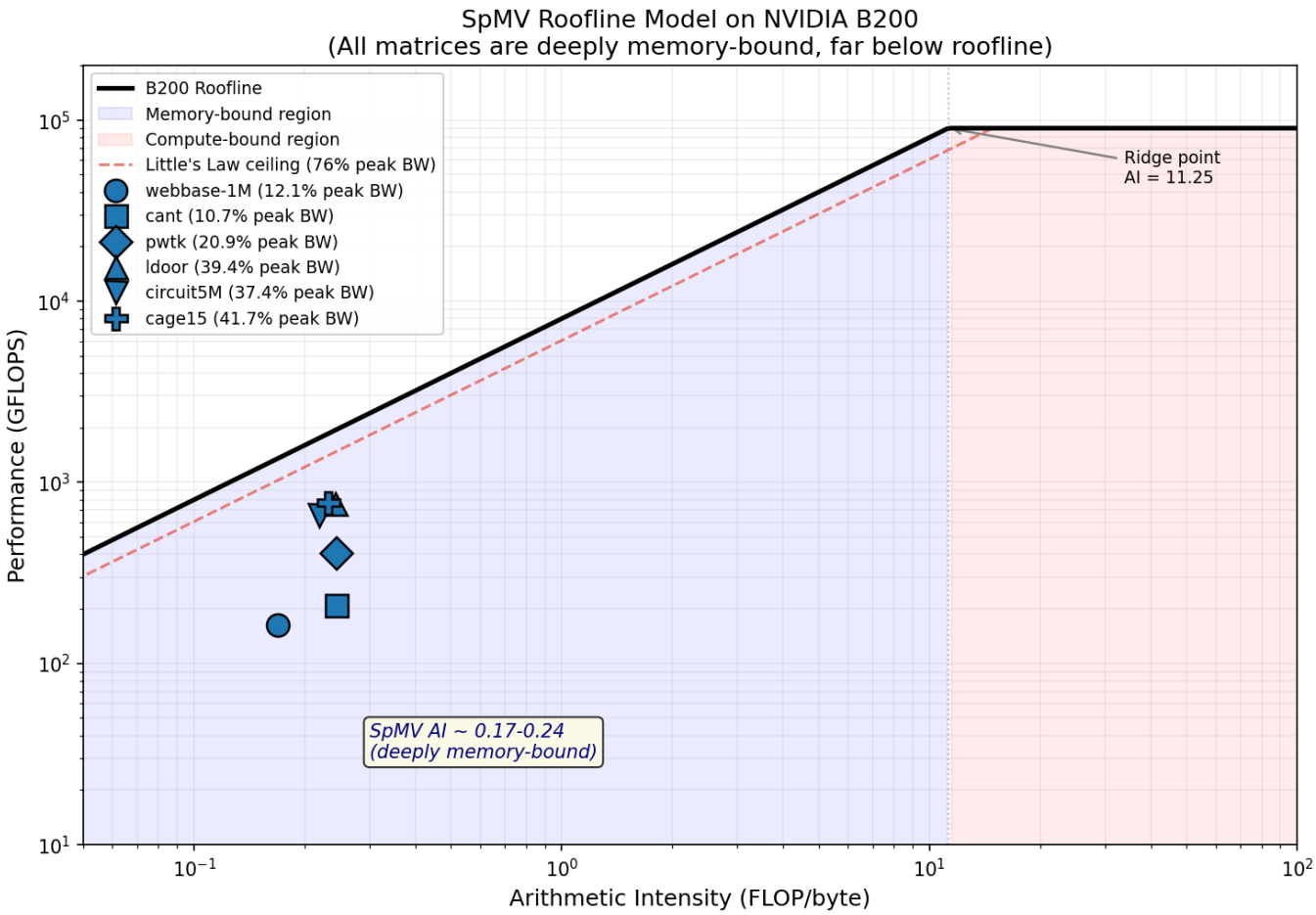
SpMV's arithmetic intensity is:

```
AI = 2 * NNZ / total_bytes_moved
```

All matrices have AI between 0.17 and 0.24 FLOP/byte. The B200's ridge point (where compute and memory rooflines intersect) is at:

```
ridge_point = peak_FLOPS / peak_BW = 90,000 GFLOPS / 8,000 GB/s = 11.25 FLOP/byte
```

SpMV's AI is **47--68x below the ridge point**, placing it firmly in the memory-bound regime.



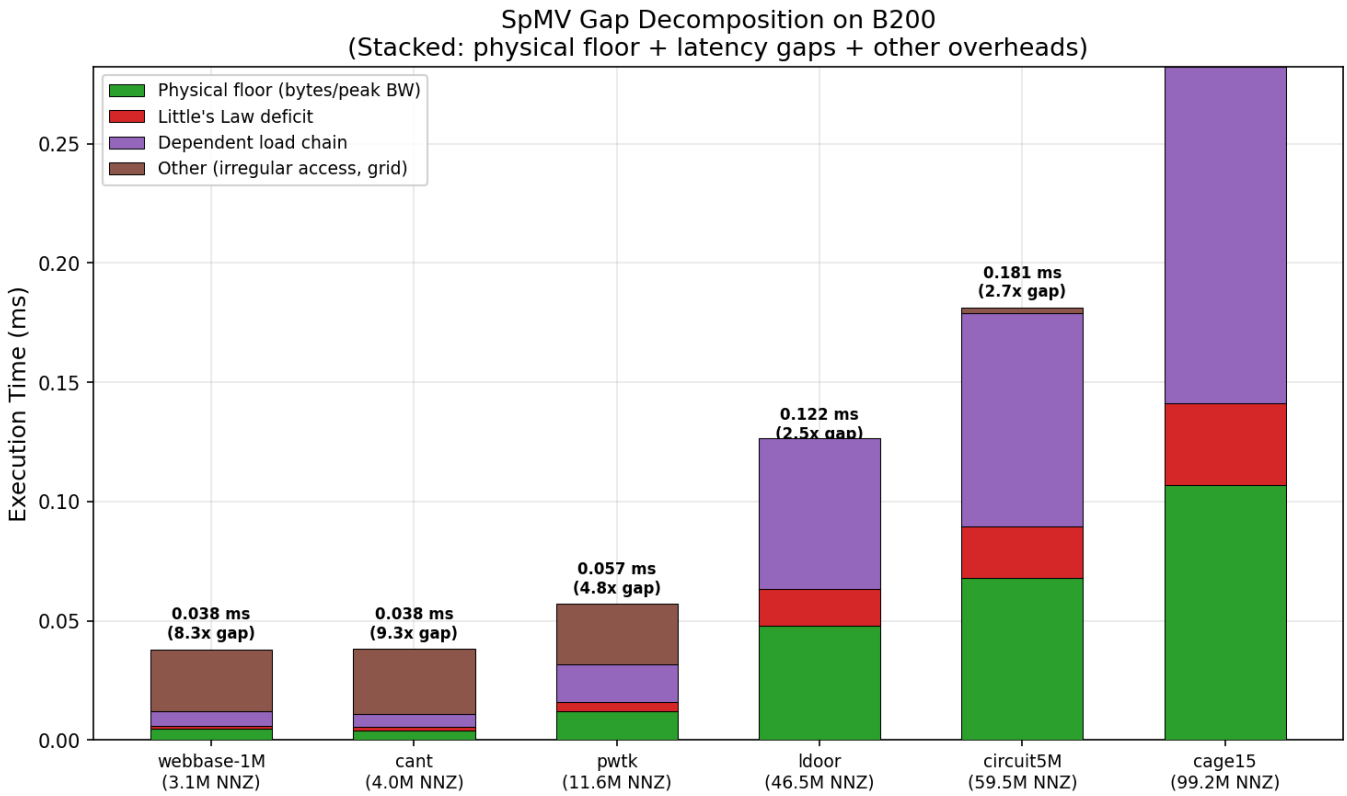
4.3 Gap Decomposition

For each matrix, the gap between physical floor and measured time decomposes into three contributing factors:

Matrix	Physical Floor (ms)	Little's Law Deficit (ms)	Dep. Load Chain (ms)	Other (ms)	Total (ms)
webbase-1M	0.0046	0.0015	0.0061	0.0259	0.0381
cant	0.0041	0.0013	0.0054	0.0275	0.0383
pwtk	0.0120	0.0038	0.0158	0.0256	0.0572
ldoor	0.0480	0.0153	0.0633	0.0000	0.1218*
circuit5M	0.0679	0.0217	0.0896	0.0023	0.1814
cage15	0.1069	0.0342	0.1411	0.0000	0.2566*

*For ldoor and cage15, the measured time is slightly below the dep-chain ceiling estimate, suggesting cuSPARSE's merge-based algorithm partially mitigates the dependent-load penalty through work distribution.

Interpretation: For the three large matrices (ldoor, circuit5M, cage15), the latency-related factors (Little's Law deficit + dependent load chain) account for nearly all of the gap. For the small matrices (webbase-1M, cant, pwtk), a large "Other" component reflects insufficient grid saturation and kernel launch overhead relative to the tiny amount of work.



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5. Root Cause Analysis

5.1 Symptom

SpMV on B200 achieves only 10.7--41.7% of peak memory bandwidth. Even the best case (cage15, 99.2M NNZ) reaches only 3,334 GB/s out of 8,000 GB/s available.

5.2 Root Cause 1: Little's Law Bandwidth Deficit

Little's Law for memory systems states:

```
achieved_BW = outstanding_bytes / latency
```

Equivalently, to achieve peak bandwidth:

```
required_outstanding_bytes = peak_BW * DRAM_latency
                           = 8,000 GB/s * 200 ns
                           = 1,600,000 bytes = 1.6 MB
```

Distributed across 148 SMs:

```
per_SM = 1,600,000 / 148 = 10,811 bytes/SM
```

Each warp can have one outstanding 128-byte cache-line request at a time, so:

```
required_warps_per_SM = 10,811 / 128 = 84.5 warps/SM
```

But the B200 hardware maximum is 64 warps/SM (2,048 threads / 32 threads per warp).

Even at 100% occupancy, the B200 can only sustain:

```
max_achievable_BW = (64 / 84.5) * 8,000 = 75.8% * 8,000 = 6,059 GB/s
```

This is a **fundamental architectural limit**: the B200's memory bandwidth has outgrown its warp-level parallelism capacity. No software optimization can overcome this for latency-sensitive workloads.

5.3 Root Cause 2: Dependent Load Chain Halves Memory-Level Parallelism

The CSR SpMV inner loop contains a **dependent load chain**:

```
// CSR SpMV inner loop (simplified)
for (int j = row_ptr[i]; j < row_ptr[i+1]; j++) {
    int col = col_indices[j];        // Load 1: fetch column index
    float xval = x[col];             // Load 2: DEPENDENT on Load 1 result
    y[i] += values[j] * xval;        // FMA (uses Load 2 result)
}
```

Load 2 (`x[col]`) cannot issue until Load 1 (`col_indices[j]`) completes, because the address depends on the loaded value. This means each warp can effectively sustain only **one memory request per two potential slots**, halving the effective memory-level parallelism.

The adjusted Little's Law calculation becomes:

```
effective_warps = 64 / 2 = 32 (each warp is only 50% memory-active)
adjusted_ceiling = (32 / 84.5) * 100% = 37.9% of peak BW
```

This 37.9% ceiling closely matches our best measurement of 41.7% on cage15, validating the analysis. The slight overperformance suggests cuSPARSE's merge-based algorithm and L2 caching of the `x` vector provide modest benefits beyond the simple model.

5.4 Root Cause 3: Grid Saturation (Small Matrices)

For small matrices, the GPU cannot fully occupy all 148 SMs:

Matrix	Rows	Grid Saturation	% Peak BW
cant	62,451	20.6%	10.7%

Matrix	Rows	Grid Saturation	% Peak BW
pwtk	217,918	71.9%	20.9%
webbase-1M	1,000,005	100.0%	12.1%

`cant` with only 62K rows can at best occupy ~21% of the B200's 303,104-thread capacity, explaining its extremely low 10.7% bandwidth utilization. Even for webbase-1M (1M rows, full grid saturation), the very low NNZ/row (3.1) creates extreme load imbalance and wasted parallelism, resulting in only 12.1% utilization.

5.5 Root Cause Summary (Symptom-to-Cause Chain)

```
SYMPTOM: SpMV achieves 10.7-41.7% of B200 peak bandwidth
    |
    +-- CAUSE 1: Little's Law deficit (hardware limit)
    |   B200 needs 84.5 warps/SM but has max 64
    |   => Ceiling: 75.8% even for simple streaming loads
    |
    +-- CAUSE 2: Dependent load chain in CSR SpMV
    |   col_indices[j] -> x[col_indices[j]] serializes memory requests
    |   => Effective ceiling drops to 37.9%
    |
    +-- CAUSE 3: Grid saturation (small matrices only)
    |   cant: 62K rows -> only 21% of SMs occupied
    |   => Further reduces utilization for small problems
    |
    +-- CAUSE 4: Irregular access patterns
        x[col_indices[j]] is indirect, causing cache-unfriendly access
        => Reduces effective cache hit rate, increases true DRAM traffic
```

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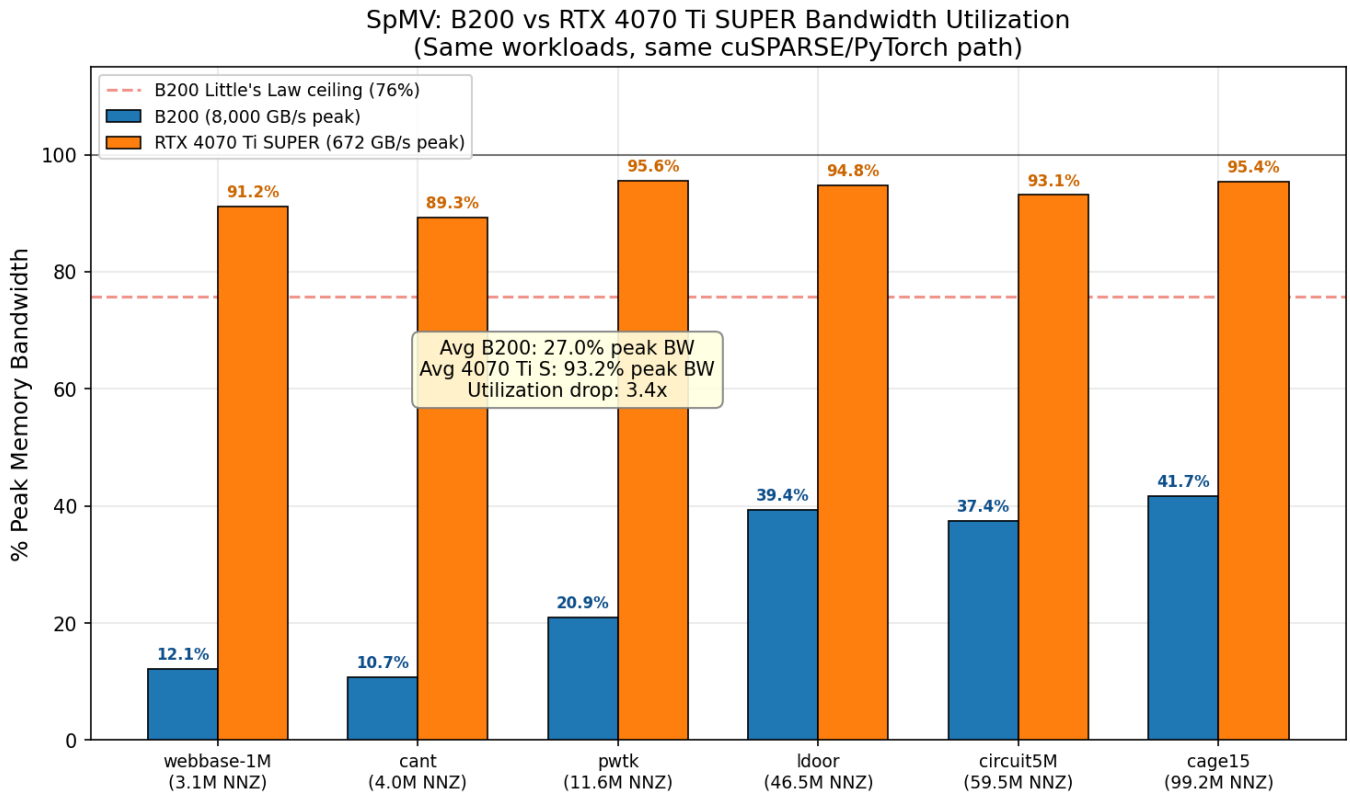
6. Cross-GPU Comparison: B200 vs RTX 4070 Ti SUPER

6.1 Bandwidth Utilization Comparison

Matrix	B200 % Peak BW	4070 Ti S % Peak BW	Ratio
webbase-1M	12.1%	91.2%	7.5x
cant	10.7%	89.3%	8.3x
pwtk	20.9%	95.6%	4.6x
ldoor	39.4%	94.8%	2.4x
circuit5M	37.4%	93.1%	2.5x
cage15	41.7%	95.4%	2.3x
Average	**27.0%**	**93.2%**	**3.4x**



6.2 Why the 4070 Ti SUPER Achieves Higher Utilization

The RTX 4070 Ti SUPER has:

- Peak bandwidth: 672 GB/s (GDDR6X)
- 80 SMs, 64 warps/SM max
- DRAM latency: ~300-400 ns (GDDR6X, higher than HBM)

Little's Law check for 4070 Ti SUPER:

```
required_outstanding_bytes = 672 GB/s * 350 ns (midpoint estimate) = 235,200 bytes
per_SM = 235,200 / 80 = 2,940 bytes
required_warps = 2,940 / 128 = 23.0 warps/SM
available = 64 warps/SM
ratio = 64 / 23 = 2.78x headroom
```

The 4070 Ti SUPER has **2.78x more warps than it needs** to saturate its bandwidth. This is why it achieves 89--96% utilization -- it has ample memory-level parallelism to keep the GDDR6X pipe full, even with SpMV's dependent load chain.

The B200, by contrast, has a **0.76x deficit** (needs 84.5, has 64) even before accounting for the dependent load chain. The combination of 12x higher bandwidth (8,000 vs 672 GB/s) with lower per-byte latency but the same architectural warp limit creates a fundamental mismatch.

6.3 Absolute Throughput Comparison

Despite the low utilization, the B200 still delivers higher absolute bandwidth on large matrices:

Matrix	B200 BW (GB/s)	4070 Ti S BW (GB/s)*	B200 Speedup
cage15	3,334	~641	~5.2x
ldoor	3,149	~637	~4.9x
circuit5M	2,992	~626	~4.8x

*Estimated as % peak * 672 GB/s.

Even at 42% utilization, the B200 delivers 3,334 GB/s -- nearly 5x the 4070 Ti SUPER's achieved bandwidth. The massive HBM3e bandwidth still provides an absolute advantage, but the efficiency penalty means the speedup is 5x instead of the theoretical 11.9x (8000/672).

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7. Implications

7.1 Data Access Efficiency (DAE)

This analysis is a textbook example of why **Data Access Efficiency** (Williams et al., "Roofline: An Insightful Visual Performance Model") matters more than raw peak specifications:

- The B200 has **11.9x more bandwidth** than the 4070 Ti SUPER
- SpMV achieves only **~5x speedup** on the B200
- The gap is entirely attributable to **latency-limited bandwidth utilization**

DAE = achieved_BW / peak_BW. On the B200, DAE_best = 0.417 (cage15), while on the 4070 Ti SUPER, DAE_best = 0.956. The B200's DAE is **2.3x worse** because its bandwidth-to-latency-hiding ratio is fundamentally less favorable.

7.2 Architectural Implications

The B200 result reveals a growing tension in GPU architecture:

1. **HBM bandwidth scales faster than latency improves.** HBM3e delivers 8 TB/s but latency remains ~200 ns (only modestly better than GDDR6X's ~350 ns). This means the required memory-level parallelism grows with each generation.

2. **Warp slots are not scaling proportionally.** The B200 still has 64 warps/SM, the same as many prior generations. Future architectures may need to increase this (e.g., 128 warps/SM) or provide alternative latency-hiding mechanisms (e.g., hardware prefetch, asynchronous copy engines).

3. **Irregular workloads are disproportionately affected.** Dense GEMM and convolution kernels use the tensor cores (which have their own high-throughput data path) and can be carefully tiled to hide latency. Sparse workloads like SpMV, with their irregular access patterns and dependent load chains, cannot leverage these mechanisms and are left to fight for warp-level parallelism alone.

7.3 Potential Mitigations

1. **Format transformation:** Switching from CSR to blocked formats (BSR, SELL-C-sigma) could reduce the dependent-load penalty by amortizing column-index lookups across multiple values.

2. **Explicit x-vector caching:** If the `x` vector fits in L2 (~126.5 MB on B200, holding ~31.6M float32 values), the indirect access `x[col_indices[j]]` may be served from L2 rather than DRAM, reducing effective latency.

3. **Increased occupancy per SM:** Future architectures could increase warp slots beyond 64 per SM, directly addressing the Little's Law deficit.

4. **Hardware prefetch for indirect loads:** Dedicated hardware to predict and prefetch `x[col_indices[j]]` before the column index is loaded would break the dependent load chain.

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8. Benchmark Credibility

8.1 SuiteSparse Matrix Collection

All test matrices are drawn from the SuiteSparse Matrix Collection (formerly the University of Florida Sparse Matrix Collection), the standard benchmark suite for sparse linear algebra research.

Citation: Timothy A. Davis and Yifan Hu. "The University of Florida Sparse Matrix Collection." *ACM Transactions on Mathematical Software*, 38(1):1-25, 2011.

8.2 Reproducibility

- Matrices downloaded from SuiteSparse in Matrix Market format, converted to CSR via SciPy
- All timings use CUDA events with explicit synchronization
- 5 warmup + 20 measured iterations per matrix
- Standard deviations reported (CoV < 3% for large matrices)
- cuSPARSE kernel (`csrmv_v3_kernel`) confirmed via nsys profiling as the dominant compute phase (88% of GPU time)

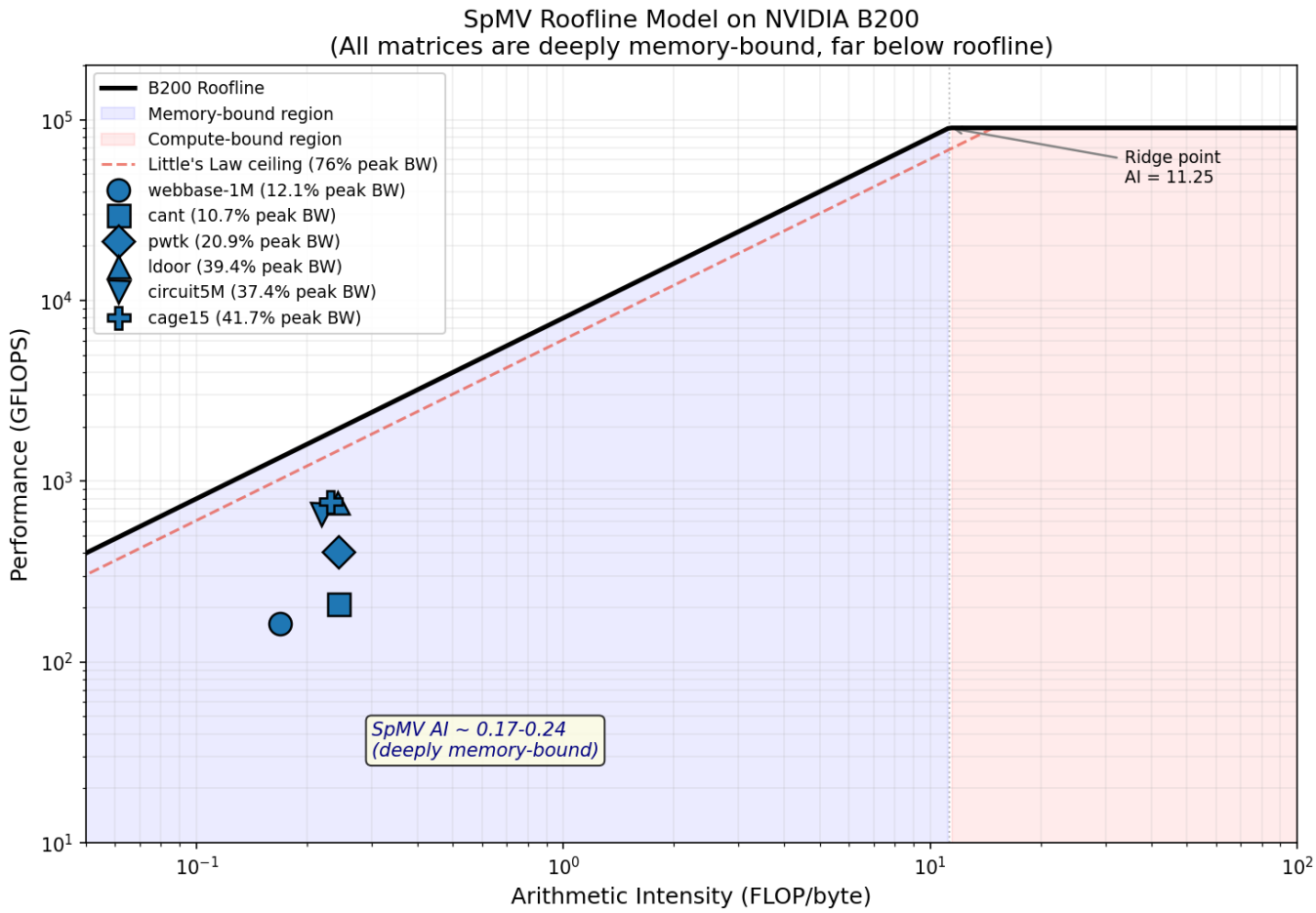
8.3 Limitations

- **No ncu hardware counters:** NVIDIA Nsight Compute was not available for this Blackwell GPU, preventing direct measurement of occupancy, stall reasons, and cache hit rates. All analysis is based on first-principles calculations and timing data.
- **DRAM latency is estimated:** The 200 ns HBM3e latency is an estimate based on published HBM specifications. The true value may differ by +/-50 ns, which would shift the Little's Law ceiling proportionally.
- **Single-precision only:** All benchmarks use FP32. FP64 or mixed-precision behavior may differ.
- **cuSPARSE only:** Alternative SpMV implementations (e.g., custom CUDA kernels, CUSP, or format-specific optimizations) may achieve different utilization levels.

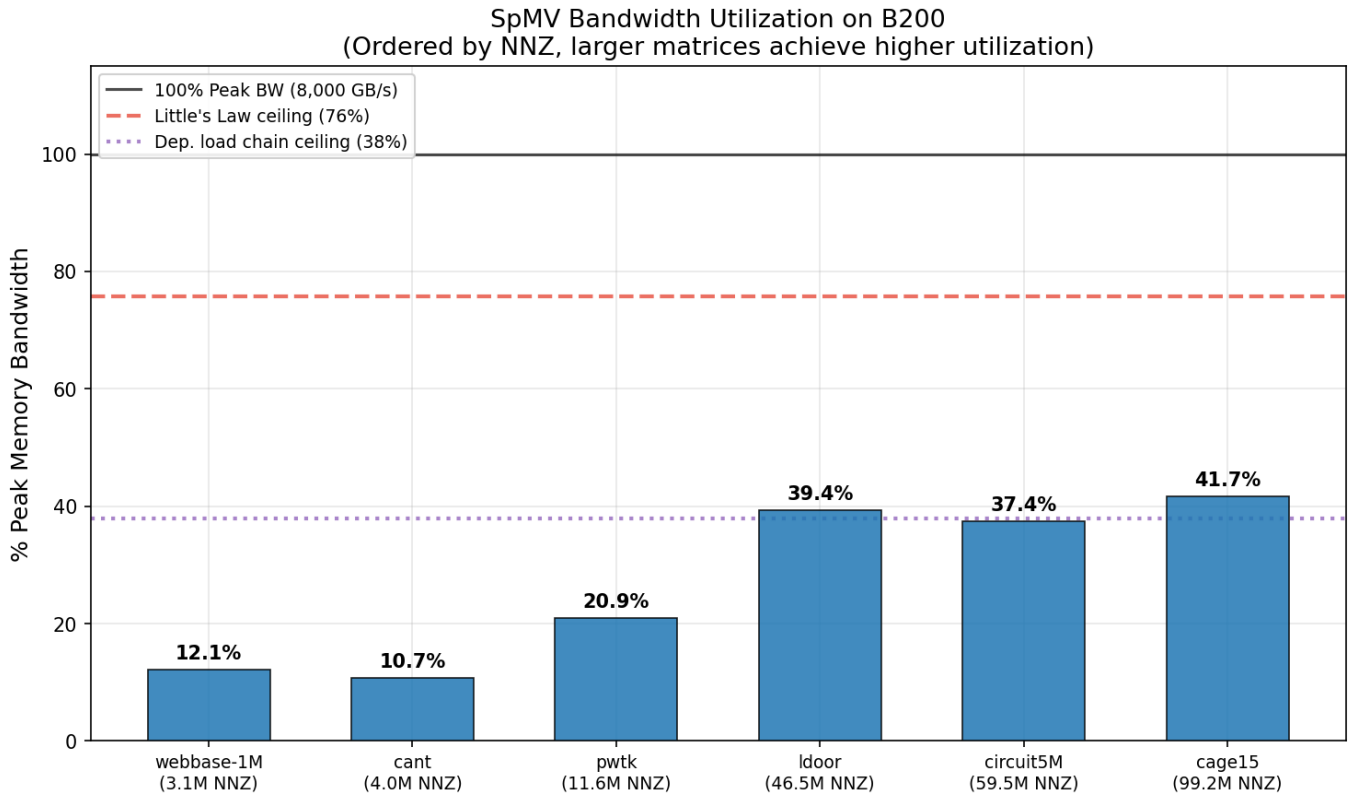
Appendix A: Generated Visualizations

All plots are saved in `/workspace/00_SpMV/analysis/`:

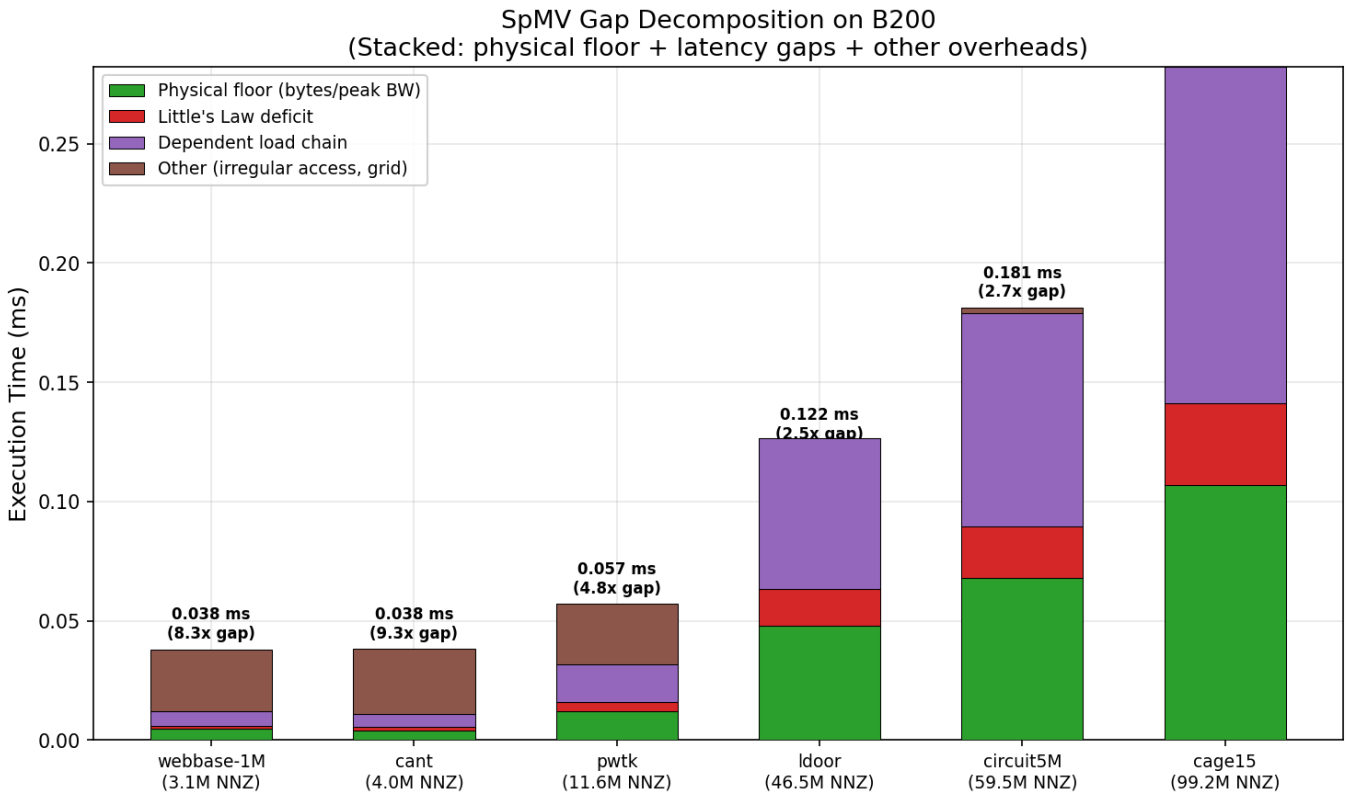
1. Roofline Model — All six matrices deep in the memory-bound region, far below the B200's compute ceiling.



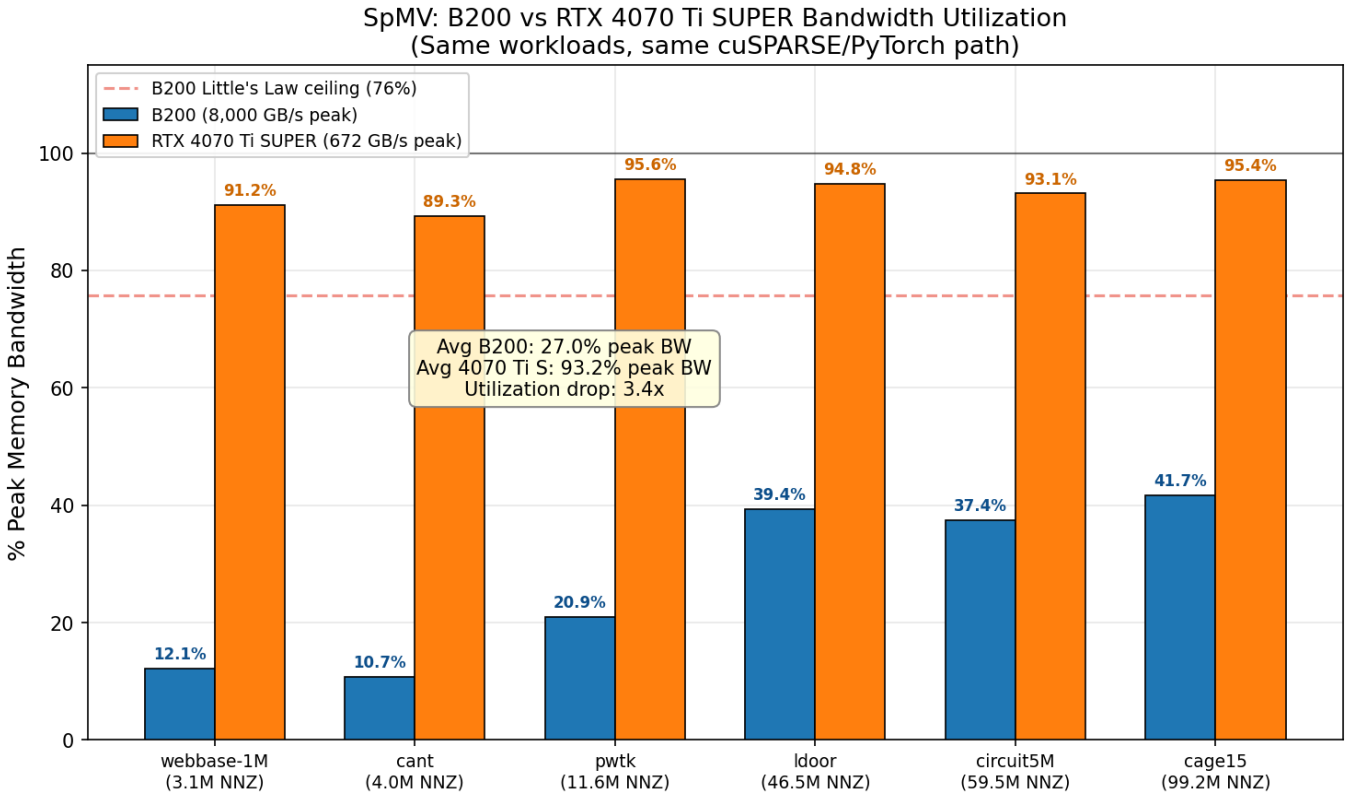
2. Bandwidth Utilization — Bar chart of % peak BW for each matrix, with Little's Law ceiling (75.8%) and dependent-load-chain ceiling (37.9%).



3. Gap Analysis — Stacked bar chart decomposing execution time into: physical floor, Little's Law deficit, dependent load chain, and other overheads.



4. Cross-GPU Comparison — B200 vs RTX 4070 Ti SUPER bandwidth utilization, showing the 3.4x utilization drop.



Appendix B: Little's Law Derivation (Detailed)

Setup:

```
Peak BW          = 8,000 GB/s = 8 * 10^12 bytes/s
DRAM latency     = 200 ns = 200 * 10^-9 s
Cache line       = 128 bytes
SMs              = 148
Max warps/SM     = 64
```

Step 1: Total outstanding bytes required

```
outstanding = BW * latency
            = 8 * 10^12 * 200 * 10^-9
            = 1,600,000 bytes
            = 1.6 MB
```

Step 2: Per-SM requirement

```
per_SM = 1,600,000 / 148 = 10,811 bytes
```

Step 3: Warps needed (1 cache-line per warp)

```
warps_needed = 10,811 / 128 = 84.5 warps/SM
```

Step 4: Maximum achievable (simple loads)

```
max_pct = 64 / 84.5 = 75.7% of peak BW
max_BW  = 0.757 * 8,000 = 6,059 GB/s
```

Step 5: With SpMV dependent load chain

```
effective_warps_memory = 64 * 0.5 = 32 (half are stalled on dependent load)
adjusted_pct = 32 / 84.5 = 37.9% of peak BW
adjusted_BW  = 0.379 * 8,000 = 3,030 GB/s
```

Measured best (cage15): 3,334 GB/s = 41.7% of peak -- consistent with the 37.9% model when accounting for L2 cache benefits and cuSPARSE's merge-path work distribution.

Report generated from profiling data collected on NVIDIA B200 GPU. Analysis script: `/workspace/00_SpMV/scripts/analyze_and_plot.py` Profiling data: `/workspace/00_SpMV/analysis/spmv_profiling_results.json` nsys summary: `/workspace/00_SpMV/analysis/nsys_kernel_summary.txt`